P_Q: let $\mathbf{V} = \bigoplus_i V_i$, let $p_i : \mathbf{V} \to V_i \to \mathbf{V}$ be the projection onto V_i, and for any path $p = h_1 \dots h_m$ let $a_p = x_{h_1} \dots x_{h_m} : V_{h'_m} \to V_{h''_1}$ be the composition of the operators corresponding to the edges occurring in p (and the action of this operator on the other V_i is zero).

It is clear that the above assignments $\mathbf{V} \mapsto (p_i \mathbf{V})$ and $(V_i) \mapsto \bigoplus_i V_i$ are inverses of each other. Thus, we have a bijection between isomorphism classes of representations of the algebra P_Q and of the quiver Q.

Remark 2.8.7. In practice, it is generally easier to consider a representation of a quiver as in Definition 2.8.3.

We lastly define several previous concepts in the context of quiver representations.

Definition 2.8.8. A **subrepresentation** of a representation (V_i, x_h) of a quiver Q is a representation (W_i, x'_h) where $W_i \subseteq V_i$ for all $i \in I$ and where $x_h(W_{h'}) \subseteq W_{h''}$ and $x'_h = x_h|_{W_{h'}} : W_{h'} \longrightarrow W_{h''}$ for all $h \in E$.

Definition 2.8.9. The **direct sum** of two representations (V_i, x_h) and (W_i, y_h) is the representation $(V_i \oplus W_i, x_h \oplus y_h)$.

As with representations of algebras, a nonzero representation (V_i) of a quiver Q is said to be irreducible if its only subrepresentations are (0) and (V_i) itself, and it is said to be indecomposable if it is not isomorphic to a direct sum of two nonzero representations.

Definition 2.8.10. Let (V_i, x_h) and (W_i, y_h) be representations of the quiver Q. A **homomorphism** $\varphi : (V_i) \longrightarrow (W_i)$ of quiver representations is a collection of maps $\varphi_i : V_i \longrightarrow W_i$ such that $y_h \circ \varphi_{h'} = \varphi_{h''} \circ x_h$ for all $h \in E$.

Problem 2.8.11. Let A be a $\mathbb{Z}_+$-graded algebra, i.e., $A = \bigoplus_{n \geq 0} A[n]$, and $A[n] \cdot A[m] \subset A[n+m]$. If $A[n]$ is finite dimensional, it is useful to consider the Hilbert series $h_A(t) = \sum \dim A[n] t^n$ (the generating function of dimensions of $A[n]$). Often this series converges to a rational function, and the answer is written in the form of such a function. For example, if $A = k[x]$ and $deg(x^n) = n$, then

$$h_A(t) = 1 + t + t^2 + \dots + t^n + \dots = \frac{1}{1-t}.$$